## Creating MavenAnsibleWebApp1

**Step 1:**creates a new ansible web application project named **MavenAnsibleWebApp1** using the **maven-archetype-webapp** code.

mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp1 -DarchetypeArtifactId=maven-archetype-webapp -DinterativeMode=false

```
shreya@ShreyaM:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp1 -DarchetypeArtifactId=maven-archetype-webapp -Din
teractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shreya/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MavenAnsibleWebApp1
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shreya/devops/MavenAnsibleWebApp1
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.080 s
[INFO] Finished at: 2026-04-15T09:49:36+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops$
```

### Step 2: Configure Maven Ansible Web App Project (pom.xml)

In this step, we move to the project directory using cd MavenAnsibleWebApp and open the Maven configuration file using gedit pom.xml. Then, we write the required project details, dependencies, WAR packaging settings, and build configuration for creating the web application.

```
shreya@ShreyaM:~/devops$ cd MavenAnsibleWebApp1
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$ ls
pom.xml  src
Show Applications
aM:~/devops/MavenAnsibleWebApp1$ gedit pom.xml
```

```
<projectxmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/mavenv4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MavenAnsibleWebApp1</artifactId>
<packaging>war</packaging>
<version>1.0-SNAPSHOT</version>
<name>MavenAnsibleWebApp1</name>
<url>http://maven.apache.org</url>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
<!--  Servlet API for Java EE  -->
<dependency>
<groupId>javax.servlet</groupId>
<artifactId>javax.servlet-api</artifactId>
<version>4.0.1</version>
```

```
<scope>provided</scope>

</dependency>

</dependencies>

<build>

<plugins>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-war-plugin</artifactId>

<version>3.4.0</version>

<!--  Update to latest stable version  -->

</plugin>

</plugins>

<finalName>MavenAnsibleWebApp1</finalName>

<!--  create a web page with the name MavenAnsibleWebApp, if we want some other name we can chage it  -->

</build>

</project>
```

**Step 3: Create Ansible Configuration Files**

In this step, we create a new folder named ansible inside the project directory using mkdir ansible. Then, we move into the folder using cd ansible and create the required Ansible files using gedit hosts.ini and gedit playbook.yml. These files are used to define target hosts and deployment tasks for automation.

```
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$ mkdir ansible
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$ ls
ansible  pom.xml  src
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$ cd ansible
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ ls
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ gedit hosts.ini
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ ls
hosts.ini
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ gedit playbook.yml
Killed
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ ls
hosts.ini  playbook.yml
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ gedit playbook.yml
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ ls
hosts.ini  playbook.yml
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ cd ..
```

**gedit host.ini**

This command is written inside hosts.ini file to define the Ansible inventory. It creates a group named web and adds localhost as the target system, using local connection and Python 3 interpreter for executing Ansible tasks.

**[web] localhost ansible_connection=local**

**ansible_python_interpreter=/usr/bin/python3**

**gedit playbook.yml**

This code is written inside playbook.yml file to automate deployment using Ansible. It targets the web host group, copies the generated WAR file from the Jenkins workspace to the Tomcat webapps folder, and then restarts the Tomcat service so the updated application runs successfully.

```
- name: Deploy to Tomcat   hosts: web   become: yes

  tasks:
```

- name: Copy WAR file to Tomcat

copy:

src:

"/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1CICD/tar

get/MavenAnsibleWebApp1.war" dest:

"/opt/tomcat/webapps/MavenAnsibleWebApp1.war" remote_src:

no

- name: Stop Tomcat

shell: /opt/tomcat/bin/shutdown.sh

ignore_errors: yes

- name: Start Tomcat

shell: /opt/tomcat/bin/startup.shStep 4:

**Build the Maven Ansible Web Application**

In this step, we run the command mvn clean install to clean previous files and build the project.

```
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MavenAnsibleWebApp1 >-------------------
[INFO] Building MavenAnsibleWebApp1 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MavenAnsibleWebApp1 --
-
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MavenAnsib
leWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i
.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
```

```
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MavenAnsibleWebApp1 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MavenAnsibleWebApp1 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MavenAnsibleWebApp1] in [/home/shreya/devops/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/shreya/devops/MavenAnsibleWebApp1/src/main/webapp]
[INFO] Building war: /home/shreya/devops/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MavenAnsibleWebApp1 ---
[INFO] Installing /home/shreya/devops/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war to /home/shreya/.m2/repository/com/example/MavenAnsibleWebApp1/
1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.war
[INFO] Installing /home/shreya/devops/MavenAnsibleWebApp1/pom.xml to /home/shreya/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsib
leWebApp1-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  38.474 s
[INFO] Finished at: 2026-05-09T23:30:54+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$
```

## Step 5: Verify Project Structure

In this step, we run the tree command to display the complete project structure.

```
.
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── Jenkinsfile
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── MavenAnsibleWebApp1
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    ├── MavenAnsibleWebApp1.war
    └── maven-archiver
        └── pom.properties

14 directories, 10 files
```

\

**Setting Up Jenkins Pipeline for MavenAnsibleWebApp**

**Step 1:** To deploy **MavenAnsibleWebApp**, open the terminal, cd into the **MavenAnsibleWebApp** folder, create a **Jenkinsfile** containing the pipeline code, and then commit and push it to GitHub.

```
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1/ansible$ cd ..
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$ gedit Jenkinsfile
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$ ls
ansible  Jenkinsfile  pom.xml  src  target
shreya@ShreyaM:~/devops/MavenAnsibleWebApp1$
```

**Edit pom.xml:**

pipeline {  agent any  // Use any

available agent  environment {

LANG = 'en_US.UTF-8'

LC_ALL = 'en_US.UTF-8'

}  // this has to be added only if you get an error saying UTF required is 8 but showing in ISO00009

tools {  maven 'Maven'  // Ensure this matches the name configured

in Jenkins

}  stages {

stage('Checkout') {

steps {

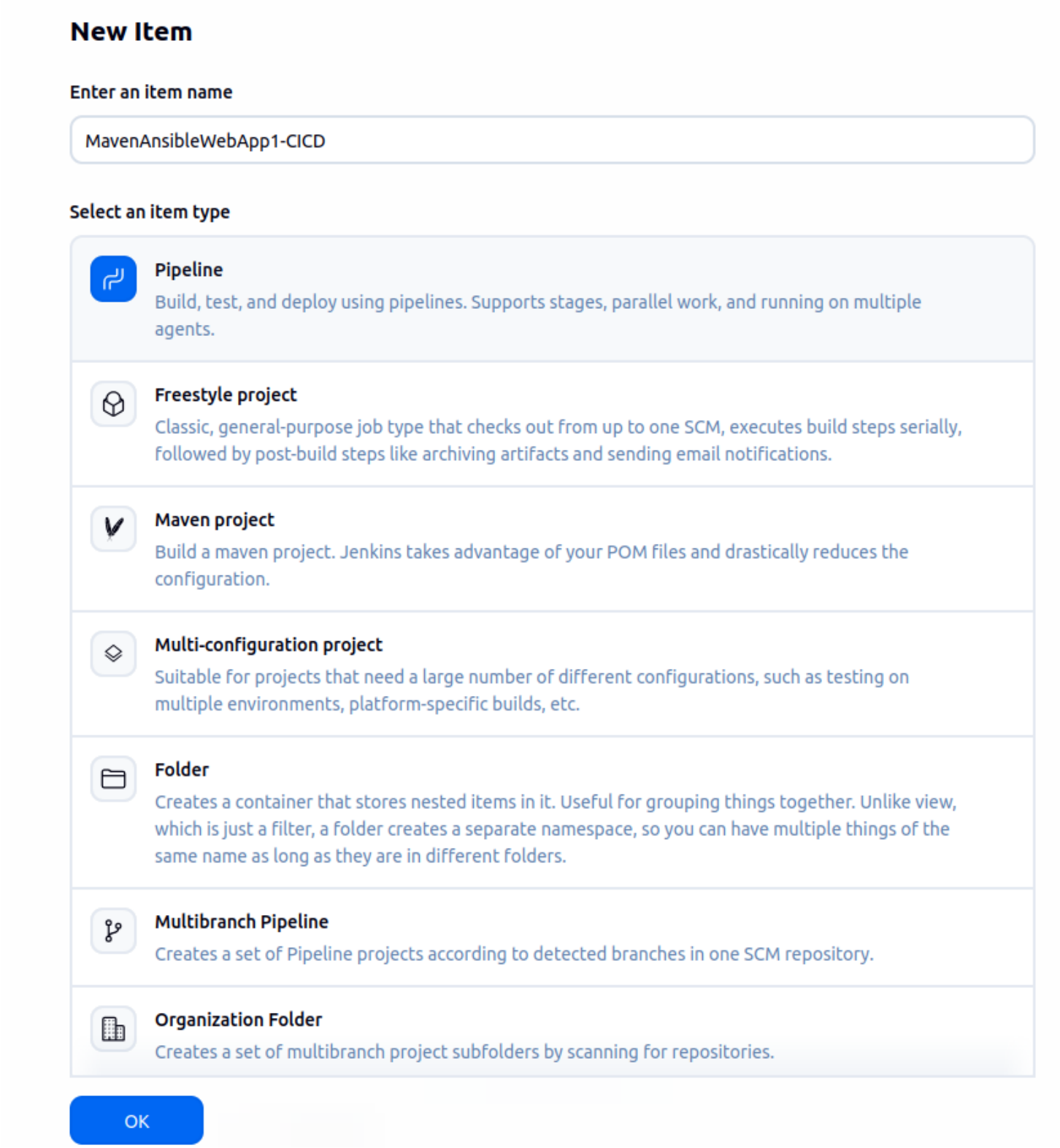
git branch: 'master', url: 'https://github.com/shreya20m /MavenAnsibleWebApp1CICD.git'

}

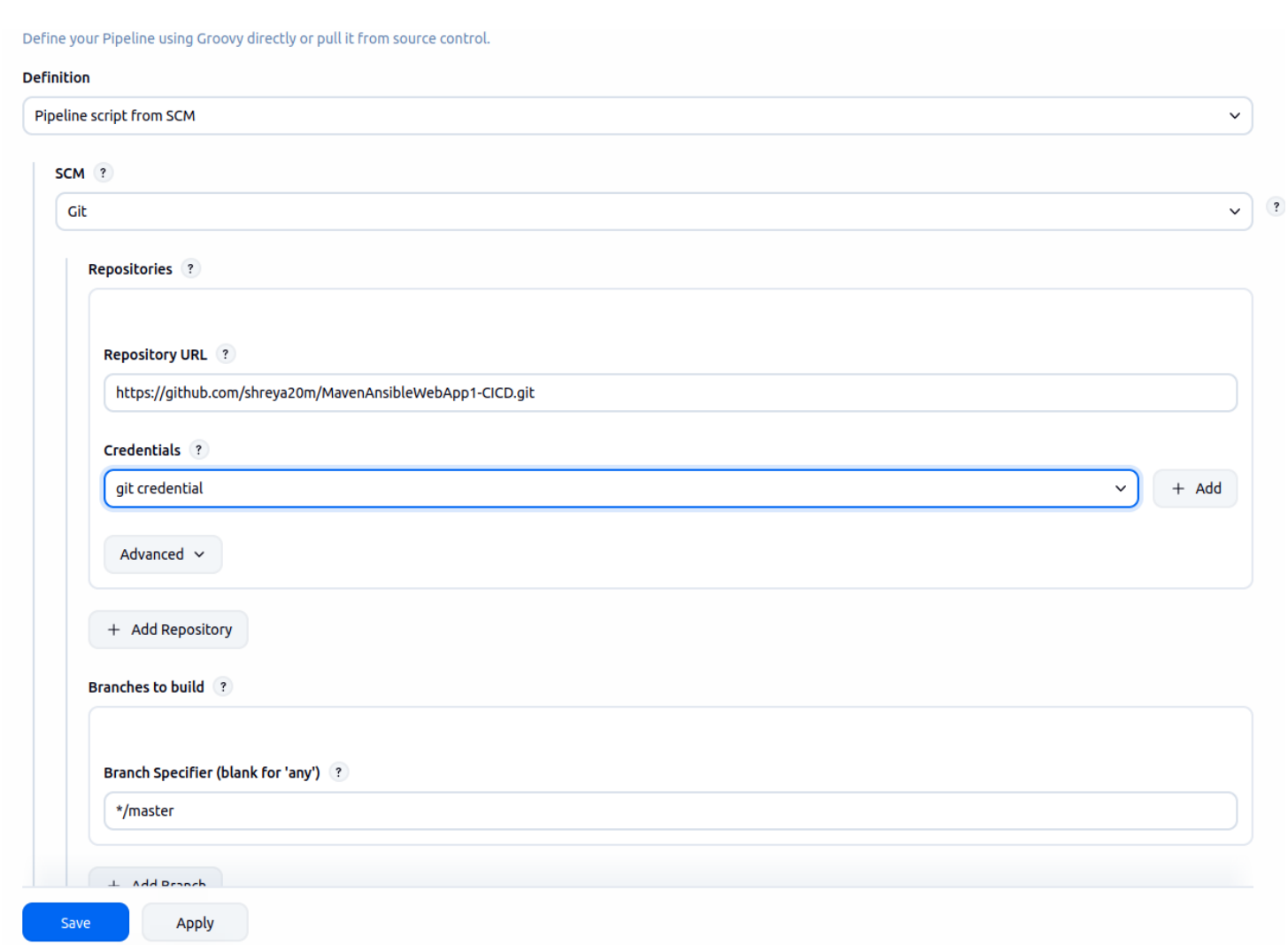
}

```
        stage('Build') {
 steps {  sh 'mvn clean package'  // Run Maven build
            }
        }
     stage('Archive') {
 steps {  archiveArtifacts artifacts: 'target/*.war', fingerprint:true
            }
        }
        stage('Deploy') {
 steps {  sh 'mvn clean package'
sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
            }
        }
    }
   }
```

**Step 2: Jenkinsfile:** To deploy **MavenAnsibleWebApp1**.go to the Jenkins home page and click on **"New Item"** enter the item name and select **Pipeline** as the project type. Click **"OK"** .



**Step 3:** In **MavenAnsibleWebApp**, go to the **Pipeline** section and under **Definition**, select **Pipeline script from SCM**. Choose **Git** as the SCM, and under **Repository URL**, paste the GitHub URL of **MavenAnsibleWebApp**. In the **Credentials** field, select the previously created credential (e.g., *GitHub Credential*). Finally, in **Script Path**, enter the name of the Jenkinsfile you created in the MavenAnsibleWebApp folder and pushed to GitHub. Then click **Apply** and **Save** to finalize the configuration.



## Step 4: Build in Jenkins

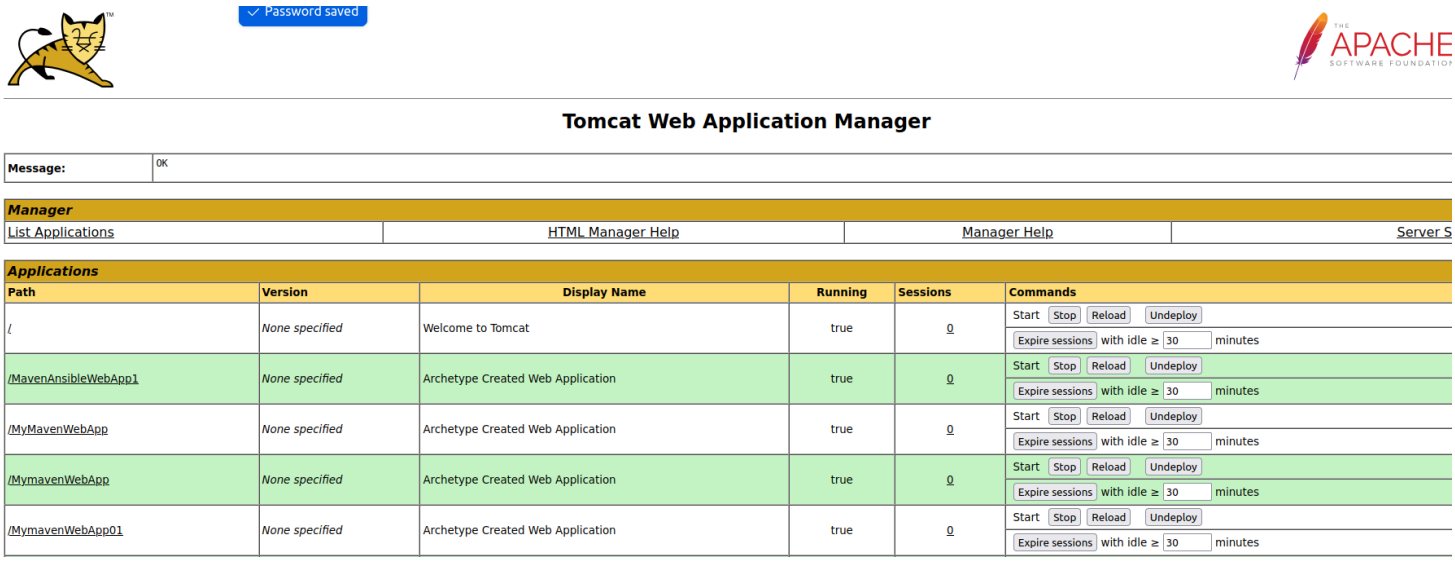
After clicking **Apply and Save**, go back to MavenAnsibleWebApp1-CICD in Jenkins and click **Build Now**. And resolve the issue.

**Step 5:** After making the changes, go back to **MavenAnsibleWebApp1-CICD** in Jenkins and click **Build Now**. The build should complete successfully without any errors.

| | Declarative: Checkout SCM | Declarative: Tool Install | Checkout | Build | Archive | Deploy |
|---|---|---|---|---|---|---|
| Average stage times: (full run time: ~49s) | 4s | 2s | 3s | 20s | 2s | 23s |
| #5 May 10 14:28 1 commit | 2s | 553ms | 2s | 10s | 662ms | 26s |
| #4 May 10 14:24 1 commit | 2s | 422ms | 3s | 18s | 1s | 15s failed |
| #3 May 10 14:14 1 commit | 6s | 6s | 5s | 28s | 4s | 21s failed |
| #2 May 10 14:04 1 commit | 3s | 600ms | 2s | 21s | 936ms | 23s failed |
| #1 May 10 13:58 No Changes | 4s | 2s | 2s | 24s | 4s | 30s failed |

**Step 6 :** Verify MavenAnsibleWebApp Deployment on Tomcat

start the server using sh startup.sh. After Tomcat starts successfully, we open the browser and access http://localhost:9090 to verify the deployed application output.



Tomcat Web Application Manager

| Message: | OK |
|---|---|

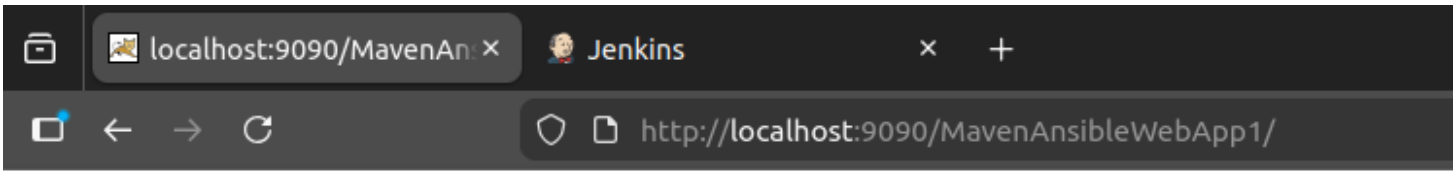
**Manager**

| List Applications | HTML Manager Help | Manager Help | Server S |
|---|---|---|---|

**Applications**

| Path | Version | Display Name | Running | Sessions | Commands |
|---|---|---|---|---|---|
| / | *None specified* | Welcome to Tomcat | true | 0 | Start Stop Reload Undeploy<br>Expire sessions with idle ≥ 30 minutes |
| /MavenAnsibleWebApp1 | *None specified* | Archetype Created Web Application | true | 0 | Start Stop Reload Undeploy<br>Expire sessions with idle ≥ 30 minutes |
| /MyMavenWebApp | *None specified* | Archetype Created Web Application | true | 0 | Start Stop Reload Undeploy<br>Expire sessions with idle ≥ 30 minutes |
| /MymavenWebApp | *None specified* | Archetype Created Web Application | true | 0 | Start Stop Reload Undeploy<br>Expire sessions with idle ≥ 30 minutes |
| /MymavenWebApp01 | *None specified* | Archetype Created Web Application | true | 0 | Start Stop Reload Undeploy<br>Expire sessions with idle ≥ 30 minutes |

Click on the project and to find the output.

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